

UNIVERSITY ERP SYSTEM - TEST PLAN & SUMMARY

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1. INTRODUCTION & SCOPE

This document outlines the testing strategy used to verify the University ERP System. The objective was to ensure the Desktop Application correctly interacts with the Cloud Database, enforces business rules (deadlines, capacity), and maintains strict role-based security.

The testing scope covers:

- Functional Testing: Verifying core features for Students, Instructors, and Admins.
- Negative Testing: Ensuring invalid actions (e.g., over-capacity) are blocked.
- Security Testing: Verifying authentication, role isolation, and account lockouts.
- Maintenance Mode: Verifying the global system lock.

2. TEST ENVIRONMENT & DATA

All tests were performed using the sample data generated by the DatabaseSeeder.java utility.

A. USER ACCOUNTS

Role: Admin
Username: admin
Password: admin123
Notes: Full system access.

Role: Instructor
Username: inst1
Password: pass123
Notes: Dept: Computer Science. Teaches CS101.

Role: Student
Username: student1
Password: pass123
Notes: Auto-enrolled in CS101.

Role: Student
Username: student2
Password: pass123
Notes: New student (no enrollments).

B. COURSE DATA

Course: CS101
Instructor: inst1
Schedule: Mon/Wed 10:00
Constraints: Cap: 50. Deadlines active.

Course: ECE201
Instructor: inst1
Schedule: Tue/Thu 14:00
Constraints: Cap: 40.

3. ACCEPTANCE TESTS (FUNCTIONAL)

The following tests verify the core functional requirements mandated by the project brief.

A. AUTHENTICATION MODULE

ID: AUTH-01

Test Case: Valid Login

Steps: Login as 'student1' with correct password.

Expected: Dashboard opens successfully.

Result: PASS

ID: AUTH-02

Test Case: Invalid Password

Steps: Login as 'admin' with 'wrongpass'.

Expected: Error message "Incorrect username or password."

Result: PASS

ID: AUTH-03

Test Case: Role Mismatch

Steps: Login as 'student1' but select "Instructor" from dropdown.

Expected: Error message "Role mismatch. Please select Student."

Result: PASS

B. STUDENT MODULE

ID: STU-01

Test Case: View Catalog

Steps: Navigate to "Course Catalog".

Expected: Table lists CS101/ECE201 with "Seats" and "Status" columns.

Result: PASS

ID: STU-02

Test Case: Register

Steps: Register for an OPEN section.

Expected: Success message. Course appears in "My Registrations".

Result: PASS

ID: STU-03

Test Case: View Timetable

Steps: Navigate to "Timetable".

Expected: Registered class appears in the weekly grid.

Result: PASS

ID: STU-04

Test Case: Drop Section

Steps: Click "Drop Selected" on a registered course.

Expected: Success message. Course removed from lists.

Result: PASS

ID: STU-05

Test Case: Transcript

Steps: Click "Download CSV".

Expected: File saves containing course codes and grades.

Result: PASS

C. INSTRUCTOR MODULE

ID: INST-01

Test Case: My Sections

Steps: Login as 'inst1'.

Expected: Table shows only CS101 and ECE201.
Result: PASS

ID: INST-02
Test Case: Enter Grades
Steps: Enter 85 for Student1 (Quiz).
Expected: "Success: Score saved."
Result: PASS

ID: INST-01
Test Case: Compute Final
Steps: Click "Compute Finals" for Student1.
Expected: Calculates weighted average (20/30/50 rule) and assigns Letter Grade.
Result: PASS

D. ADMIN MODULE

ID: ADM-01
Test Case: Create User
Steps: Create new Student profile via dashboard.
Expected: Success. New user can log in immediately.
Result: PASS

ID: ADM-01
Test Case: Create Section
Steps: Schedule a new section with Deadlines.
Expected: Section appears in Student Catalog.
Result: PASS

ID: ADM-01
Test Case: System Logs
Steps: View "System Logs" tab.
Expected: Shows login history and failed attempts.
Result: PASS

4. EXTRA & NEGATIVE TESTS (CUSTOM LOGIC)

These tests cover specific validation logic and edge cases implemented to ensure data integrity.

A. BUSINESS RULES

Test Case: Late Registration
Logic: Deadline Check
Steps: Admin sets Reg Deadline to yesterday. Student tries to register.
Expected: Error "Registration deadline passed." Status shows "CLOSED".
Result: PASS

Test Case: Full Capacity
Logic: Capacity Check
Steps: Admin sets Cap=1. One student registers. Second student tries.
Expected: Error "Section is full (1/1)." Status shows "FULL".
Result: PASS

Test Case: Duplicate Entry
Logic: Integrity Check
Steps: Student tries to register for the same course twice.
Expected: Error "Already registered."
Result: PASS

B. SECURITY & MAINTENANCE

ID: SEC-01

Test Case: Cross-Grading

Logic: Access Control

Steps: Instructor tries to grade a section they don't teach.

Expected: Blocked: "Security Warning! You can only grade your own sections."

Result: PASS

ID: SEC-02

Test Case: Account Lock

Logic: Brute Force Protection

Steps: User enters wrong password 5 times.

Expected: Error "Account locked! Try again in 5 minutes." Logs updated.

Result: PASS

ID: MAINT-01

Test Case: Global Lock

Logic: Maintenance Mode

Steps: Admin toggles Maintenance ON. Student tries to register.

Expected: Blocked: "System is in maintenance." Banner appears on UI.

Result: PASS

5. AUTOMATED UNIT TESTS

In addition to manual testing, the following automated JUnit 5 tests were implemented to verify logic in isolation.

StudentServiceTest.java

- testRegister_WhenMaintenanceOn_ShouldFail (Passed)
- testRegister_WhenDeadlinePassed_ShouldFail (Passed)
- testRegister_WhenSectionFull_ShouldFail (Passed)
- testRegister_Success (Passed)

AdminServiceTest.java

- testCreateStudent_Success (Passed)
- testDeleteSection_WhenStudentsEnrolled (Passed - Ensures data integrity)

InstructorServiceTest.java

- testComputeFinalGrade (Passed - Verifies 20/30/50 weighting logic)
- testSecurityCheck (Passed - Verifies instructors cannot grade unassigned sections)

6. TESTING SUMMARY

Total Tests Executed: 23

Passed: 23

Failed: 0

Conclusion:

The system meets all functional and non-functional requirements. The application handles edge cases (full classes, deadlines) gracefully and maintains data integrity during concurrent operations.